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Machine learning in biology

Week 1: and an introduction to foundation models

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This lecture

Roughly two hours, with two short videos and one interactive activity.

- Part 1 Welcome and module orientation
- Part 2 Why machine learning for biology, healthcare and nutrition
- Part 3 What machine learning is: the vocabulary and the workflow
- Part 4 From a generic example to a biological one
- Part 5 Foundation models, conceptually
- Part 6 Cost, sustainability and choosing the right size of model

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Welcome and module orientation

Part 1 · What this module is, and how it works

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Your lecturer

- Dr Haithem Afli, Department of Computer Science, MTU
- Principal Investigator, national ADAPT Centre; ADAPT lead at MTU
- Lead of the Human-Centred AI research group
- Research: natural language processing, trustworthy and explainable AI, machine learning for biology and the social sciences
- Chair of the Computer Science postgraduate research board
- IEEE Senior Member; ACM and ACL member
- Several themes in this module, particularly Week 12 on trust and evaluation, come directly out of that research

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The module

Code and title	COMP9085 Machine Learning in Biology
Credits and level	5 credits, NFQ Level 9 (MSc)
Contact	2 h lecture + 2 h laboratory each week
Independent study	About 3 h each week
Assessment	Coursework only. Project 1 (50%) and Project 2 (50%)
Examination	None
Reassessment	Resubmitted coursework only
Descriptor	courses.cit.ie · moduleId 15193

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Learning outcomes

Unchanged from the approved module descriptor.

- LO1 Analyse machine learning workflows to facilitate pre-processing, dimensionality reduction and model selection
- LO2 Compare and apply appropriate machine learning algorithms to specific biological datasets
- LO3 Develop supervised machine learning models for classification tasks
- LO4 Develop unsupervised machine learning models for clustering tasks
- LO5 Evaluate the accuracy of predictive models using standard methods and interpret their results

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Assessment

Unchanged from the approved descriptor. Marks are for the reasoning, not for the tool: using a foundation model earns nothing on its own.

	Weight	What it asks of you
Project 1: supervised biological machine learning	50%	Build, tune and honestly evaluate a supervised model on biological or clinical data
Project 2: unsupervised biological machine learning	50%	Find structure in unlabelled biological data, and show it is not an artefact
Written examination	None	The module is assessed entirely by coursework

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How the parts fit together

One copy of the teaching material, linked from Canvas, rather than two copies that drift apart.

Each week

- Lecture, two hours. Slides on Canvas and the website beforehand, as PowerPoint and PDF
- Website: concepts, worked example, an interactive figure you manipulate, a self-check
- Laboratory, two hours, from Week 2
- One recommended reading, chosen because it is the most useful thing to read at that point

Where things live

- Teaching material: the course website
- Announcements, submission and grades: Canvas
- Slides: both, as PPTX and PDF
- Your weekly check-in: your own browser, private to you

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There is no laboratory in Week 1.

The first practical is in Week 2, once you have the decision-tree material to work with. Before then: install Python, watch the two videos, read Greener et al. (2022). None of it is assessed.

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Software and resources

Retained from the original resources slides.

- Python 3 with NumPy, pandas, scikit-learn, Matplotlib, seaborn
- Jupyter for the labs
- Nothing in this module needs a GPU
- Install before Week 2 so the first lab starts with the material
  - pip install numpy pandas scikit-learn matplotlib seaborn jupyter
- UCI Machine Learning Repository, CDC NHANES, openFDA and the NCI Genomic Data Commons are the main data sources
- The full list, with links, is on the resources page of the website

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Why machine learning for biology, healthcare and nutrition

Part 2 · The data, and why it is difficult

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The biological data landscape

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Recognising which kind of problem you have decides everything downstream.

Five families of data, five different problems

Genomics
long discrete sequences

→

Proteomics
sequence with 3-D consequences

→

Medical imaging
high-dimensional pixel grids

→

Clinical records
sparse, irregular, confounded

→

Nutrition
self-reported, high error

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Why these domains are harder than the usual benchmarks

None of this is a reason not to do it. All of it is a reason to evaluate carefully.

- Small n, large p. Hundreds of samples, tens of thousands of features
- Labels are expensive, and are often a clinician's judgement rather than a measurement
- Rows are not independent: several samples come from one patient, one site, one batch
- Measurement error is substantial and systematic, not random noise
- The two kinds of error almost never cost the same
- Populations are unevenly represented, and the gaps are rarely recorded
- A model that is accurate and unusable is a common outcome

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Try this interactively

Watch: Biology Meets Computer Science

What is it about modern biological data that makes computation necessary rather than merely convenient?

`Quanta Magazine · 3 min 46 s · on the Week 1 page`

Open the week page on the course website and scroll to Interactive exploration.

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What machine learning is

Part 3 · Vocabulary, workflow, generalisation

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Machine learning

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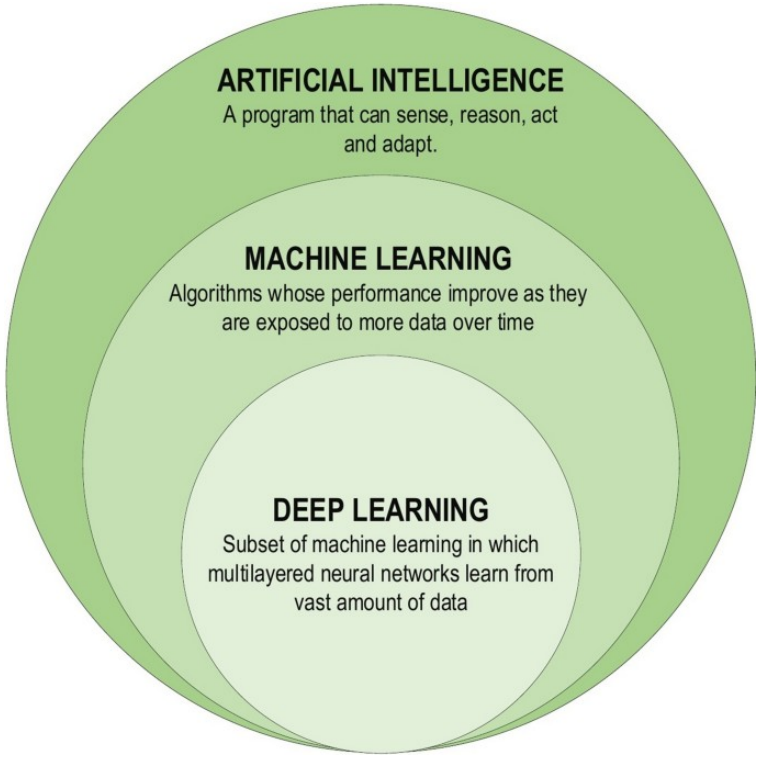
Fitting a function from data rather than specifying it by hand.

The data, not the programmer, determines the rule. Everything else in this module is a consequence of that one sentence.

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Where machine learning sits

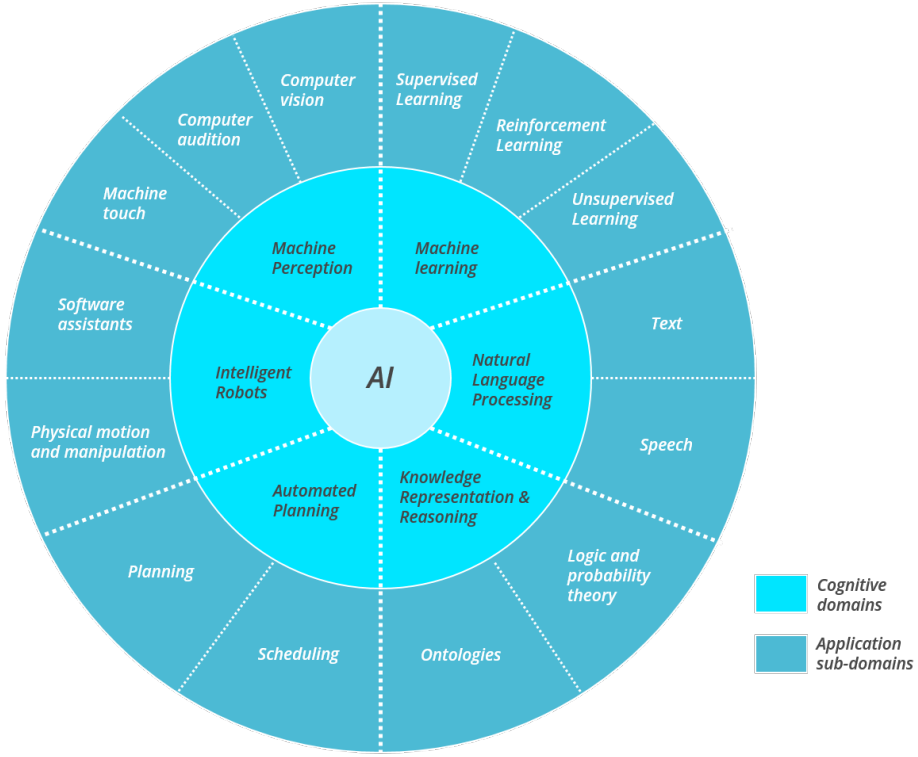


Artificial intelligence contains machine learning, which contains deep learning. Foundation models are deep learning at a scale that changes what is practical, not a fourth ring.

Figure retained from the 2025/26 COMP9085 Week 1 deck (slide 11).

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Artificial intelligence: the domains



Machine learning is one cognitive domain of AI among several. This module is the inner segment marked machine learning, and the application sub-domains it reaches in biology.

Figure retained from the 2025/26 COMP9085 Week 1 deck (slides 4 and 11).

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Step 1: building a machine-learning solution

$$D = [(d_1, t_1), (d_2, t_2), \ldots, (d_n, t_n)]$$
$$M = Learn(D)$$

A training dataset D pairs each instance description d with its target value t. A learning algorithm consumes D and returns a model M.

- **d** the descriptive features of one instance: the measurements you have
- **t** the target feature: what you are trying to predict
- **D** the training dataset, n such pairs
- **M** the fitted model, the output of learning

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Step 2: deploying a machine-learning solution


```
t = M(q)
```

A query instance q, described by the same features, is passed to the fitted model, which returns a prediction t.

- **q** a new instance the model has never seen
- **M** the model fitted in step 1
- **t** the prediction: a class, or a value

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The vocabulary, precisely

- Feature (descriptive feature): an input column
- Label (target feature): what you are predicting
- Training data: the examples the algorithm learns from
- Model: the function produced by learning
- Prediction: the model applied to a new instance
- Generalisation: performance on data the model has never seen
- Evaluation: measuring generalisation honestly, which is harder than it sounds and is the subject of Weeks 3 and 12

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The three task types you will build

LO3 is the first row, LO4 is the third. The second is supervised learning with a continuous target and is treated alongside it.

Task	The label is	A biological example	Weeks
Classification	one of a fixed set of classes	Is this breast-mass biopsy benign or malignant?	2-6, 8, 10-11
Regression	a number on a continuous scale	Predict a patient's HbA1c from dietary recall data	3, 10
Clustering	not given at all	Which tumour subtypes are present in this expression matrix?	7, 9

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Generalisation is the only thing that matters.

A model that reproduces its training data perfectly and fails on a new patient has learned nothing useful. Week 3 measures this properly; Week 12 asks whether the measurement can be believed.

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Categories of machine learning

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Retained from slides 41 to 44, with the examples made biological.

By supervision

- Supervised: every example is labelled. Classification predicts a category, regression predicts a quantity
- Unsupervised: no labels. Find structure
- Semi-supervised: a little labelled data, a lot unlabelled. The common situation in biology
- Reinforcement: learning from interaction and reward. Not covered in this module

By what you are asking for

- Classification: benign or malignant
- Regression: serum vitamin D in nmol/L
- Clustering: are there subtypes in these tumours?
- Dimensionality reduction: 20,000 genes into a few useful axes
- Representation learning: learn the features rather than choose them

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Prediction is not explanation

A model can predict an outcome accurately while encoding no causal mechanism.

The two goals need different evidence. Conflating them is the most common error in applied biological machine learning, and it is the error this module works hardest to prevent.

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Try this interactively

What type of ML problem is this?

Eight real biological scenarios. Classify each one, then read why the tempting wrong answer is tempting.

`weeks/week01.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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From a generic example to a biological one

Part 4 · Why we start with loans and finish with tumours

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A small, deliberately generic example

Six applicants. Can you write a rule that separates those who repay from those who default?

Applicant	Occupation	Age	Loan	Loan-salary ratio	Outcome
Pierre Lundi	professional	36	20,000	3.22	?
Astrid Karlsson	professional	41	15,000	4.64	?
Mary Smyth	industrial	34	50,000	2.96	?
Heidi Fischer	industrial	61	5,000	2.52	?
Klaus Haller	professional	37	60,000	1.51	?
Luis Lopez	industrial	33	6,000	5.25	?

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Now make it twenty-five rows and seven features.

Do not attempt this by hand. The original slide carries the instruction: 'don't actually ask them to do this one - the point here is that this is hard!' That difficulty is the entire argument for machine learning.

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Now change the columns and keep the structure

The machine-learning problem is identical. Everything around it is not.

The generic example	The biological version
Loan applicant	Breast-mass biopsy from one patient
Occupation, age, loan-salary ratio	Nuclear radius, texture, concavity: 30 measurements
Repaid or defaulted	Benign or malignant
A wrong decision costs money	A false negative is a missed cancer
Data is abundant and cheaply labelled	Labels need a pathologist or a five-year follow-up
Nobody asks who is in the dataset	Who is in the dataset determines who the model works for

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The algorithms are the easy part.

Where the measurements came from, how reliable they are, who is in the data, what each error costs, and what you would have to show before anyone should act on the output. That is the module.

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Feature engineering

Every method in Weeks 2 to 6 takes engineered features as given. Week 7 asks where they come from.

What it is deciding, by hand, which numbers describe the object well enough for a model to work with: amino-acid composition, molecular weight, nuclear radius, mean daily energy intake

Why it works a good feature encodes real domain knowledge, so a small model trained on little data can still succeed

Why it is the bottleneck the features fix a ceiling on what any downstream model can learn; no algorithm recovers information the features threw away

Who does it someone who knows the biology, which makes it slow, expensive and hard to transfer to a new problem

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Two proteins can have identical composition and opposite function.

Amino-acid composition is order-blind by construction. Any model built on it alone is at chance on any question that depends on order. Week 7 runs this comparison on real data and reports both numbers.

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Representation learning

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Letting the model learn the description instead of specifying it.

The model is trained on a very large amount of unlabelled biological data first, with no task in mind, and learns a vector representation of each sequence. That vector is then used as the input to an ordinary classifier. Nobody chose what goes into it.

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Foundation models, conceptually

Part 5 · What they are, and where they fit

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Try this interactively

Watch: AlphaFold, the making of a scientific breakthrough

What information must a model learn about proteins before it can predict a structure at all? Where would that information come from?

`Google DeepMind · 7 min 54 s · on the Week 1 page`

Open the week page on the course website and scroll to Interactive exploration.

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AlphaFold, read carefully

A striking result and a careful reading of its limits are not in tension. Both are required.

- Predicting 3-D protein structure from sequence was open for fifty years
- AlphaFold 2 produced predictions competitive with experiment for many CASP14 targets (Jumper et al., Nature 2021)
- AlphaFold DB now covers over 214 million protein sequences, openly (Varadi et al., NAR 2024)
- It reports its own confidence as pLDDT: above 90 very high, 70 to 90 confident, 50 to 70 low, below 50 very low
- Very low pLDDT correlates with intrinsic disorder, so it is often information rather than failure
- It does not tell you what a protein does, or which conformation matters in a cell
- It is not a conventional supervised classifier: learned representations plus strong geometric assumptions

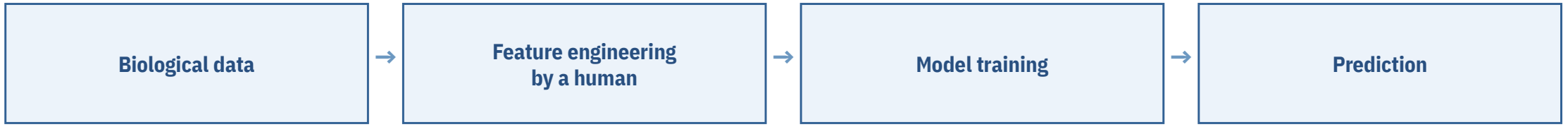
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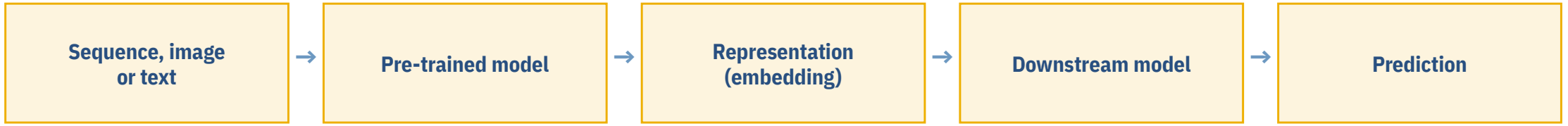
Three workflows, at a conceptual level

You are not expected to implement any of these this week.

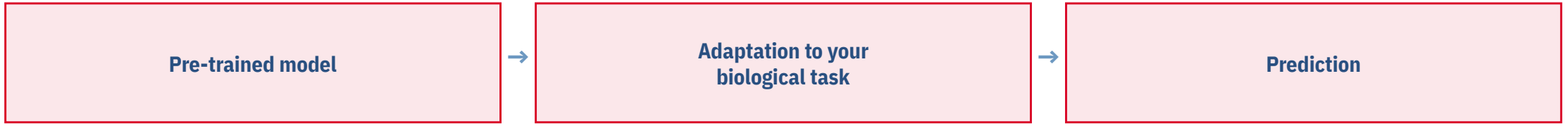
Classical machine learning



The foundation-model workflow



Fine-tuning



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Foundation model

A large model pretrained with self-supervision on a broad corpus, intended to be adapted to many downstream tasks rather than built for one.

Note what is not in the definition. Size is not: ESM-2 is released as small as 8 million parameters and is still a foundation model. Architecture is not either.

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Biological foundation models you will meet

Five names, one idea: biological input goes in, a fixed-length vector comes out, and an ordinary classifier goes on top.

Model	Trained on	Turns this into a vector	First met in
ESM-2	protein sequences (UniRef)	an amino-acid sequence	Week 7
ProtT5	protein sequences	an amino-acid sequence	Week 7
DNABERT-2	genomic DNA	a DNA sequence	Week 7
Nucleotide Transformer	genomes across many species	a DNA sequence	Week 7
HyenaDNA	genomic DNA, very long context	a long DNA sequence	Week 11

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Foundation models do not replace machine-learning fundamentals. They build on them.

This is the argument of the whole module. It is why Weeks 2 to 6 are classical machine learning taught properly, and why the foundation-model weeks come afterwards rather than instead.

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What is the simplest appropriate machine-learning approach for this biological problem, and what evidence do you have that it is appropriate?

The central question of the module. Both halves matter, and the second one is the hard part.

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Cost, sustainability and model choice

Part 6 · Why the biggest model is not the default answer

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The carbon footprint of training large models

Retained from the original deck, and promoted from a closing remark to a model-selection criterion.

- Training large models has a measurable energy and carbon cost (Strubell, Ganesh and McCallum, ACL 2019)
- The full development cycle, including architecture search and retraining, is very much larger than a single training run
- “There's a big push to scale up machine learning to solve bigger and bigger problems, using more compute power and more data. As that happens, we have to be mindful of whether the benefits of these heavy-compute models are worth the cost of the impact on the environment.” Dan Jurafsky
- Inference at scale is not free either, and is often the larger lifetime cost
- Source: Stanford HAI, “AI's carbon footprint problem”

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Measured on the dataset this module uses throughout

Real measured results: UCI Breast Cancer Wisconsin (Diagnostic), 569 samples, 5-fold cross-validation. The simplest model wins.

Model	Accuracy	Cost
Majority-class baseline	62.7%	None
Decision tree, depth 3	91.7%	A laptop, instantly
Random forest, 500 trees	95.4%	A laptop, seconds
Logistic regression	97.4%	A laptop, instantly
PCA (10) + logistic regression	97.7%	A laptop, instantly

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A bigger model is not automatically a better scientific model.

Computational and environmental cost belong beside accuracy as a model-selection criterion, not as an afterthought. Week 12 returns to this with the full trust checklist.

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Before Week 2

Next week: decision trees, entropy and information gain, on real diagnostic data.

- No lab this week. The first practical is in Week 2
- Install Python and the scientific stack
- Watch the two videos on the Week 1 page and answer the question attached to each
- Read Greener, Kandathil, Moffat and Jones (2022), “A guide to machine learning for biologists”, Nature Reviews Molecular Cell Biology
- Try the ML Problem Classifier activity on the website
- Fill in the weekly check-in; it takes a minute and it shapes the next lecture

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References and further reading

Greener, J. G., Kandathil, S. M., Moffat, L. & Jones, D. T. (2022). A guide to machine learning for biologists. Nature Reviews Molecular Cell Biology 23:40-55. doi:10.1038/s41580-021-00407-0

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Varadi, M. et al. (2024). AlphaFold Protein Structure Database in 2024. Nucleic Acids Research 52(D1):D368-D375. doi:10.1093/nar/gkad1011

Strubell, E., Ganesh, A. & McCallum, A. (2019). Energy and Policy Considerations for Deep Learning in NLP. ACL 2019. arXiv:1906.02243

Wolberg, W., Mangasarian, O., Street, N. & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic). UCI Machine Learning Repository. doi:10.24432/C5DW2B

Videos: Quanta Magazine, “Biology Meets Computer Science”; Google DeepMind, “AlphaFold: The making of a scientific breakthrough”

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Questions

Next week: Decision trees and explainable biological prediction. The first laboratory runs in Week 2.

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